

Advanced (Habanero)

Q1. What is the factored form of $x^3 + 8$?

- (A) $(x + 2)(x^2 - 2x + 4)$
- (B) $(x - 2)(x^2 + 2x + 4)$
- (C) $(x + 2)(x^2 - 4x + 4)$
- (D) $(x - 2)(x^2 + 4)$
- (E) $(x + 2)(x^2 - 4)$

Q2. Factor $27x^3 - 64$

- (A) $(3x - 4)(9x^2 + 12x + 16)$
- (B) $(3x - 4)(9x^2 - 12x + 16)$
- (C) $(3x + 4)(9x^2 - 12x + 16)$
- (D) $(3x + 4)(9x^2 + 12x + 16)$
- (E) $(3x - 8)(9x^2 + 24x + 64)$

Q3. What is $a^3 - b^3$ equal to?

- (A) $(a + b)(a^2 + ab + b^2)$
- (B) $(a - b)(a^2 - ab + b^2)$
- (C) $(a + b)(a^2 - ab + b^2)$
- (D) $(a - b)(a^2 + ab + b^2)$
- (E) $(a + b)(a^2 + 2ab + b^2)$

Q4. Factor $8x^3 + 27$

- (A) $(2x + 3)(4x^2 - 6x + 9)$
- (B) $(2x - 3)(4x^2 + 6x + 9)$
- (C) $(2x - 3)(4x^2 - 6x + 9)$
- (D) $(2x + 3)(4x^2 + 6x + 9)$
- (E) $(2x - 3)(4x^2 + 12x + 27)$

Q5. What is the factored form of $x^3 - 64$?

- (A) $(x - 4)(x^2 + 4x + 16)$
- (B) $(x - 4)(x^2 - 4x + 16)$
- (C) $(x - 4)(x^2 + 16)$
- (D) $(x + 4)(x^2 - 16)$
- (E) $(x + 4)(x^2 + 4x + 16)$

Q6. Which of the following is a factor of $x^3 + 125$?

- (A) $x + 5$
- (B) $x - 5$
- (C) $x + 4$
- (D) $x - 4$
- (E) $x - 3$

Q7. What is $x^3 + 8x^3$ equal to?

- (A) $9x^3$
- (B) $9x^2$
- (C) $9x$
- (D) $x^3 + 8$
- (E) $x^3 - 8$

Q8. Factor $125x^3 - 8y^3$

- (A) $(5x - 2y)(25x^2 + 10xy + 4y^2)$
- (B) $(5x - 2y)(25x^2 - 10xy + 4y^2)$
- (C) $(5x + 2y)(25x^2 - 10xy + 4y^2)$
- (D) $(5x + 2y)(25x^2 + 10xy + 4y^2)$
- (E) $(5x - 2y)(25x^2 + 20xy + 16y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $64x^3 + 125y^3$. Show all steps and explain your reasoning.

Q10. Find the factored form of $x^3 - 216$. Explain how you arrived at your answer.

Chef's Tips

- Tip 1: Remind students to identify and apply the sum and difference of cubes formulas
- Tip 2: Encourage students to check their work by multiplying the factors to ensure they equal the original expression
- Tip 3: Consider providing additional support or scaffolding for struggling students